

# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

## ASSESSMENT TOOL 1 – Analytical Problem Solving

|                         |                                                                                                                   |                      |                                                       |
|-------------------------|-------------------------------------------------------------------------------------------------------------------|----------------------|-------------------------------------------------------|
| <b>Course Code:</b>     | ITA0512                                                                                                           | <b>Course Title:</b> | Computer Vision                                       |
| <b>CO Assessed:</b>     | CO2 – Imitation (BL3)                                                                                             | <b>Assessment:</b>   | Assessment Tool 1 – <b>Analytical Problem Solving</b> |
| <b>Weightage:</b>       | 10% of CIA                                                                                                        | <b>Total Marks:</b>  | 100                                                   |
| <b>CO1 Statement:</b>   | <i>Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)</i> |                      |                                                       |
| <b>Student Name:</b>    | Kaviya V                                                                                                          | <b>Reg. No.:</b>     | 192421022                                             |
| <b>Section / Batch:</b> | Third Year/ B. Tech IT                                                                                            | <b>Date:</b>         | 07/08/26                                              |

### Code of Conduct

I \_\_Kaviya V/192421022\_\_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: \_\_Kaviya V\_\_

| Criteria                  | Weightage | Level 4 (Exemplary)                                                               | Level 3 (Proficient)                      | Level 2 (Developing)                       | Level 1 (Beginning)                           | Criteria                  | Marks |
|---------------------------|-----------|-----------------------------------------------------------------------------------|-------------------------------------------|--------------------------------------------|-----------------------------------------------|---------------------------|-------|
| Problem Understanding     | 20%       | Clearly interprets problem, identifies all parameters and requirements accurately | Understands problem with minor gaps       | Partial understanding; misses key elements | Misinterprets or unable to understand problem | Problem Understanding     |       |
| Method Selection          | 20%       | Selects most appropriate method/formula with strong justification                 | Selects correct method with minor errors  | Inappropriate or partially correct method  | Unable to select method                       | Method Selection          |       |
| Solution Process          | 25%       | Logical, systematic steps with correct approach throughout                        | Mostly correct steps with minor mistakes  | Disorganized steps; multiple errors        | No clear process                              | Solution Process          |       |
| Accuracy of Calculation   | 20%       | All calculations accurate; correct final answer                                   | Minor calculation errors                  | Frequent errors; incorrect final answer    | No valid calculations                         | Accuracy of Calculation   |       |
| Interpretation of Results | 15%       | Clearly interprets results with units and engineering relevance                   | Basic interpretation with minor omissions | Limited or unclear interpretation          | No interpretation                             | Interpretation of Results |       |

# 1. High-Pass Filtering

An image requires sharpening.

## (a) Interpret the need for High-Pass Filtering

High-pass filtering is an image enhancement technique used to emphasize high-frequency components such as edges, fine details, and object boundaries while suppressing low-frequency components like smooth background and gradual intensity variations.

Need:

- Enhance image sharpness.
- Highlight object boundaries.
- Improve feature extraction.
- Make edges more visible for segmentation.

## (b) Justify the filter selection

A High-Pass Filter is selected because:

- It enhances rapid intensity changes.
- It improves image clarity.
- It preserves edge information.
- It is widely used before object detection and pattern recognition.

Therefore, a High-Pass Filter is the most suitable choice for image sharpening.

## (c) Apply the High-Pass Mask

Given High-Pass Filter Mask:

$$\begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

Suppose the image window is

$$\begin{bmatrix} 20 & 20 & 20 \\ 20 & 100 & 20 \\ 20 & 20 & 20 \end{bmatrix}$$

Multiply each element with the corresponding mask value.

| Image Pixel | Mask | Multiplication |
|-------------|------|----------------|
| 20          | -1   | -20            |
| 20          | -1   | -20            |
| 20          | -1   | -20            |
| 20          | -1   | -20            |
| 100         | 8    | 800            |
| 20          | -1   | -20            |
| 20          | -1   | -20            |
| 20          | -1   | -20            |
| 20          | -1   | -20            |

**(d) Compute the Centre Value**

**Sum all products.**

$$\begin{aligned} &= (-20) + (-20) + (-20) + (-20) + 800 + (-20) + (-20) + (-20) + (-20) \\ &= 800 - 160 \\ &= 640 \end{aligned}$$

Therefore, **Centre Pixel Value = 640**

**(e) Interpret Edge Enhancement**

Since the output value (640) is much larger than the surrounding pixels,

- edges become brighter,
- fine details are highlighted,
- the image becomes sharper.

Hence, High-Pass Filtering successfully enhances edges and improves image quality.

## 2. Frequency Domain High-Pass Filtering

**Given Frequency Components:** [10,25,40,80,120]

**Assume,**

- 10,25  $\rightarrow$  Low Frequency
- 40,80,120  $\rightarrow$  High Frequency

### (a) Interpret High-Frequency Importance

High-frequency components represent

- object boundaries
- image edges
- textures
- fine details

These components are very important for image sharpening and feature extraction.

### (b) Justify High-Pass Filtering

A High-Pass Filter removes

- smooth background
- unwanted low-frequency information and preserves
- edges
- detailed structures

Hence it is used for sharpening.

### (c) Apply Filtering

**Remove low frequencies.**

Original: [10,25,40,80,120]

Remove: [10,25]

Remaining: [40,80,120]

### (d) Compute Output

Filtered Output: [40,80,120]

### **(e) Interpret Result**

After filtering,

- background becomes less dominant,
- edges become clearer,
- image appears sharper,
- high-frequency information is preserved.

Thus, High-Pass Filtering improves edge visibility.

## **3. Adaptive Thresholding**

**Given Pixel Values:** [60,80,120,140]

**Local Mean Threshold:** [T=100]

### **(a) Interpret Segmentation Challenge**

A grayscale image may have

- uneven illumination,
- shadows,
- varying brightness.

Using one global threshold may produce incorrect segmentation.

Hence adaptive thresholding is preferred.

### **(b) Justify Adaptive Thresholding**

Adaptive Thresholding computes

- a separate threshold
- for each local region.

Advantages

- Works well under non-uniform lighting.
- Improves segmentation accuracy.

- Separates object and background effectively.

### (c) Apply Local Threshold

Rule

If

Pixel >100

Output =255

Otherwise

Output =0

| Pixel | Comparison  | Binary Output |
|-------|-------------|---------------|
| 60    | $60 < 100$  | 0             |
| 80    | $80 < 100$  | 0             |
| 120   | $120 > 100$ | 255           |
| 140   | $140 > 100$ | 255           |

### (d) Compute Binary Output

[0,0,255,255]

### (e) Interpret Result

- The first two pixels belong to the background because they are below the threshold.
- The last two pixels belong to the foreground because they are above the threshold.
- Hence adaptive thresholding successfully separates the object from the background.